

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – May 2026

EXPERIENCE

Software Engineer

BU Spark!

September 2024 – December 2024

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

Boston University Engineering IT

January 2024 – July 2025

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow*

May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows
- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

High-Frequency Market Data Processor | *C++, Boost Libraries, Custom Memory Pool, Lock-Free Queues*[Month] 2026

- Engineered low-latency market data processing engine with custom memory allocators and lock-free data structures, achieving [X] ns average processing time per message (benchmark against system clock)
- Implemented multi-threaded order book reconstruction with NUMA-aware thread pinning and CPU cache optimization, supporting [X]K messages/second throughput (stress test with synthetic data generator)
- Built real-time technical indicator calculations (SMA, EMA, VWAP) using SIMD instructions and template metaprogramming, reducing computation overhead by [X]% (compare with naive implementation)

Real-Time Analytics Pipeline | *Apache Kafka, Apache Flink, InfluxDB, React Dashboard, Docker Compose*[Month] 2026

- Built event-driven analytics system processing [X]K events/minute through Kafka topics with Flink stream processing, implementing windowed aggregations and complex event pattern detection (producer load testing)
- Designed time-series data storage in InfluxDB with optimized retention policies and continuous queries, supporting [X]-day historical analysis with [X]ms query response times (benchmark with InfluxDB CLI)
- Created real-time React dashboard with WebSocket streaming and D3.js visualizations, displaying live metrics with [X]ms end-to-end latency from event ingestion to UI update (browser performance profiling)

TECHNICAL SKILLS

Blockchain Languages: Solidity, Circom, C++, JavaScript, TypeScript, SQL (PostgreSQL)

Smart Contracts & DeFi: Ethereum, ERC-20 Tokens, MetaMask Integration, Transaction Management, Economic Incentive Mechanisms

Zero-Knowledge Proofs: Circom, Groth16 Protocol, Privacy-Preserving Verification, Cryptographic Protocols

Decentralized Storage: IPFS, Distributed Storage, SHA256 Hash Verification, Thirdweb SDK

Performance & Trading Systems: High-Frequency Trading, Lock-Free Data Structures, Market Data Processing, Technical Indicators

Real-Time Processing: Apache Kafka, Apache Flink, InfluxDB, Stream Processing, WebSocket Streaming